

Engineering Mechanics

F.E. Sem. I

EVALUATION SYSTEM

	Time	Marks
Theory Exam	3 Hrs.	80
Practical Exam	–	–
Oral Exam	–	25
Term Work	–	25
Internal Assessment	–	20

SYLLABUS

Module -1 System of Coplanar Forces

Resultant of Concurrent forces, Parallel forces, Non Concurrent, Non Parallel system of forces, Moment of force about a point, Couples, Varignon's Theorem. Distributed Forces in plane.

Center of Gravity and Centroid for plane Laminas.

Module -2 Equilibrium of System of Coplanar Forces

- **Condition of equilibrium** for concurrent forces, parallel forces and Non concurrent Non Parallel general forces and Couples.
- **Types of support**, loads, Beams, Determination of reactions at supports for various types of loads on beams.
- **Analysis of plane trusses** by using Method of joints and Method of sections.(Excluding pin jointed frames)

Module -3 Forces in Space

- **Resultant of Noncoplanar force systems** : Resultant of Concurrent force system, Parallel force system and Nonconcurrent nonparallel force system.
- **Equilibrium of Noncoplanar force systems** : Equilibrium of Concurrent force system, Parallel force system and Nonconcurrent nonparallel force system.
- **Friction** : Introduction to Laws of friction, Cone of friction, Equilibrium of bodies on inclined plane, Application to problems involving wedges, ladders.

Module -4 Kinematics of Particle

Velocity & acceleration in terms of rectangular co-ordinate system, Rectilinear motion, Motion along plane curved path, Tangential & Normal component of acceleration, Motion curves (a-t, v-t, s-t curves), Projectile motion, Relative velocities.

Module -5 Kinematics of Rigid Bodies

Introduction to general plane motion, Instantaneous center of rotation for the velocity, velocity diagrams for bodies in plane motion, (up to 2 linkage mechanism)

Module -6 Kinetics of a Particle

- **Force and Acceleration** : Introduction to basic concepts, D'Alemberts Principle, Equations of dynamic equilibrium, Newton's Second law of motion.
- **Work and Energy** : Principle of Work and Energy, Law of Conservation of Energy.
- **Impulse and Momentum** : Principle of Linear Impulse and Momentum. Law of Conservation of momentum. Impact and collision.

Mumbai University Question Paper Format

- 1) Question paper will comprise of 6 questions, each carrying 20 marks.
- 2) Total 4 question need to be solved.
- 3) Question No. 1 will be compulsory and based on entire syllabus wherein sub-question of 2 to 3 marks will be asked.
- 4) Remaining questions will be mixed in nature (e.g. Suppose Q.2 has part (a) from module 3 then part (b) will be from any module other than module 3) having 15 marks each.
- 5) In question paper weightage of each module will be proportional to number of respective lecture hours as mentioned in syllabus.

Distribution of Marks

Module	Chapters	Q.1 (20)	Q.2 (20)	Q.3 (20)	Q.4 (20)	Q.5 (20)	Q.6 (20)	Weightage of marks
1	1.1 System of Coplanar forces 1.2 Center of Gravity and Centroid	04 –	06 –	– 08	– –	– –	– –	10 08
2	2.1 Equilibrium of system of coplanar forces 2.2 Types of support 2.3 Analysis of plane trusses	04 – –	08 – –	– – –	– 08 –	– – 08	– – –	12 08 08
3	3.1 Forces in space 3.2 Friction	– 04	– –	06 –	– –	– –	04 08	10 12
4	4.1 Kinematics of Particle	04	–	–	06	06	04	20
5	5.1 Kinematics of Rigid Bodies	–	–	–	06	06	–	12
6	6.1 Kinetics of a Particle: Force and Acceleration 6.2 Kinetics of a Particle: Work and Energy 6.3 Kinetics of a Particle: Impulse and Momentum	04 – –	– – 06	– 06 –	– – –	– – –	04 – –	08 06 06
Total Marks		20	20	20	20	20	20	120

